

Summary

Memory Designing and Circuit Implementation

Memory designing in computer architecture focuses on constructing a required memory capacity using available smaller memory chips. The goal is to correctly configure address buses, data buses, selection logic, and chip interconnections so that the final hardware behaves exactly as specified.

Key conclusions include:

- Memory capacity is represented as $n \times m$, where the number of locations determines the address bus width and the word size determines the data bus width.
- Chip Enable (CE) / Chip Select (CS) signals are essential for managing multiple memory chips in a single system.
- Memory expansion follows grid-based structures, increasing capacity vertically (locations) and horizontally (word size).
- ROM can be used as a high-speed arithmetic unit, such as a multiplier, by storing precomputed results.
- Active Low signals are preferred for critical control lines due to their superior resistance to noise.

1. Memory Size Representation and Bus Configuration

Memory size is expressed in the form:

Memory Size = $n \times m$

Where:

- **n** = Number of memory locations (words)
- **m** = Number of bits per location (word size)

1.1 Address Bus Calculation

If a memory has N locations, the number of address lines required is:

Address Bus Width = $\log_2(N)$

Example:

- 8 locations $\rightarrow \log_2(8) = 3$ address lines

1.2 Data Bus Calculation

The width of the data bus is equal to the word size:

Data Bus Width = m

Example:

- Word size = 4 bits $\rightarrow$ Data bus = 4 bits

1.3 Examples of Bus Requirements

Memory Size	Locations (n)	Word Size (m)	Address Bus	Data Bus
8×4	8	4 bits	3-bit	4-bit
$4K \times 16$	4,096	16 bits	12-bit	16-bit
$16K \times 4$	16,384	4 bits	14-bit	4-bit
$256K \times 8$	262,144	8 bits	18-bit	8-bit
$16M \times 16$	16,777,216	16 bits	24-bit	16-bit

2. Memory Designing Using Multiple Chips

When an exact memory chip is unavailable, multiple smaller chips are combined to achieve the desired capacity.

2.1 Chip Enable (CE) / Chip Select (CS)

Each memory chip includes a selection signal:

- **Logic 1 (Active High)** $\rightarrow$ Chip is enabled
- **Logic 0** $\rightarrow$ Chip is disabled and outputs are in high-impedance state

2.2 Memory Map vs Circuit Implementation

- **Memory Map:** Logical allocation of address ranges to memory chips
- **Circuit Implementation:** Physical wiring of address lines and selection logic to match the memory map

2.3 Incremental Design Method

To construct a larger memory (for example, 16×4) from smaller chips (8×4):

1. **Internal Selection**
 - Lower-order address lines are connected to all chips
 - Selects individual locations within a chip
2. **Chip Selection**
 - Higher-order address lines are connected to decoders or logic gates
 - Ensures only one chip is active for a given address

3. Grid-Based Memory Structures

Large memory systems are organized as grids of rows and columns.

3.1 Grid Calculations

- Number of Rows = Total locations required / Locations per chip
- Number of Columns = Total word size required / Word size per chip
- Total Chips = Rows $\times$ Columns

3.2 Decoder Usage

Decoders activate one row at a time:

- 1:2 decoder $\rightarrow$ selects 2 rows using 1 address line
- 2:4 decoder $\rightarrow$ selects 4 rows using 2 address lines

If a required decoder is unavailable:

- A 4:16 decoder can be constructed using five 2:4 decoders
 - Four provide 16 outputs
 - One selects the active decoder

4. ROM as a High-Speed Multiplier

ROM can be used to perform multiplication by storing all possible results.

4.1 Logical Basis

- Hardware multipliers require complex circuitry and multiple clock cycles
- For fixed-width inputs, the number of combinations is finite

Example:

- 4-bit × 4-bit multiplication
- Total combinations = $16 \times 16 = 256$

4.2 Implementation

- Inputs act as the address
- Stored data represents the multiplication result
- Result is obtained in one clock cycle

4.3 ROM Size Calculations

- **4-bit multiplier**
256 locations × 8-bit result = 2 Kbits
- **8-bit multiplier**
 2^{16} locations × 16-bit result = 64K × 16 bits

5. Active High vs Active Low Signals

Signal polarity affects reliability and power consumption.

5.1 Active High Signals

- Functional at Logic 1 (VCC)
- Idle at Logic 0 (0 V)
- More power-efficient when idle
- Highly susceptible to noise

5.2 Active Low Signals

- Functional at Logic 0
- Idle at Logic 1 (VCC)
- Much higher noise resistance
- Preferred for critical signals such as Chip Select
- Commonly used in high-reliability systems

6. Design Complexity and Examination Insights

Memory design problems often test conceptual depth rather than standard patterns.

6.1 Non-Standard Address Mapping

- Chip selection may use middle-order address bits
- This alters starting addresses of memory blocks

Example:

- Blocks may start at 0, 8, 16, 24 instead of 0, 1024, 2048, 3072

6.2 Address Space Requirement Rule

If required memory size lies between two powers of two:

- The higher power of two determines the number of address lines

Example:

- Required memory = 96 KB
- Lies between 64 KB and 128 KB
- Address lines required = 17

Conclusion

Memory designing is a systematic process that integrates bus calculation, chip selection, decoder logic, and signal integrity. By carefully combining smaller memory units and applying correct selection logic, complex memory systems can be built efficiently. Mastery of these principles is essential for both hardware design and competitive examinations.